Mark O'Dowd

Full-Stack Software Engineer & AI Solutions Architect

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SUMMARY

Full-Stack Software Engineer with over 7 years of experience architecting scalable, cloud-native applications. Proven track record of aligning technical execution with cost-efficiency and performance. AWS Solutions Architect and multi certified in AI (Practitioner & Business Strategist), specializing in integrating secure, production grade GenAI capabilities into enterprise applications.

TECHNOLOGY STACK

Languages: TypeScript, Python, Rust
AI: Amazon SageMaker & Bedrock, Claude, OpenAI
Infrastructure: AWS, GitHub Actions, CI/CD, Docker
Frontend: React, Next.js, Tailwind CSS
Backend: Django, Node.js, Express, Axum
Databases: PostgreSQL, SQLite, DynamoDB, MongoDB

EXPERIENCE

SENIOR FULL-STACK ENGINEER (FREELANCE/CONTRACT) - Emerald Software - *Mar 2025 to Present*

www.irishtrad.ie - Architected and deployed a high-performance, full stack educational platform for traditional Irish music instruments

- **High-Performance Architecture:** Built a dual engine architecture utilising a server-side rendered (SSR) Next.js frontend for optimal performance and a low-latency, high-throughput Rust (Axum) backend REST API
- **Cloud Infrastructure & Media:** Designed and deployed a robust AWS cloud architecture leveraging ECS & Fargate for deployment, S3 for media asset hosting, RDS for database management, and SES for reliable transaction email delivery

www.fightorflight.studio - Developed a bespoke, media focused portfolio for a Manchester-based creative studio

- **Headless CMS Integration:** Engineered a highly customizable portfolio platform leveraging Contentful CMS, empowering the client to independently manage and update media assets without developer intervention
- **Media Optimization:** Implemented adaptive bitrate streaming for high resolution video assets, significantly reducing initial page load times and improving mobile performance
- **CI/CD & Infrastructure:** Automated deployment pipelines for staging and production environments to ensure seamless delivery, and secured domain email deliverability by configuring SPF, DKIM, and DMARC protocols through Cloudflare

www.jujitsu.ie - Developed a modern, mobile-first web application to digitize the World Ju-Jitsu Federation Ireland syllabus

- **Zero-Maintenance Hosting:** Designed a serverless, highly cost-efficient hosting architecture utilizing an AWS S3 static bucket, virtually eliminating ongoing infrastructure overhead.
- **Performance Optimization:** Optimized asset delivery by implementing modern WebP image compression, significantly reducing page weight and boosting load speeds.
- **Mobile-First Design:** Engineered a responsive, lightweight, mobile-first user interface to ensure student syllabus materials are highly accessible on any device during training.

SOFTWARE ENGINEER - Dawson Andrews - *May 2024 to Mar 2025* **-** Dawson Andrews specialises in product design, strategy, and software development

- Rapidly adopted the Vue/Nuxt ecosystem to lead the end to end development of an enterprise grade web application ensuring a seamless transition from design to AWS backed production

SOFTWARE ENGINEER - Hurree - *Jan 2023 to Mar 2024* **-** Hurree is a cross platform analytics dashboard that enables teams to unify data to gain one high level overview

- Engineered user friendly analytics features using React and Django, unifying unique data sources into high-performance, real-time dashboards for cross-platform business intelligence while championing code quality and system reliability by leading technical design discussions and conducting rigorous code reviews

SOFTWARE ENGINEER - Micro Focus - *Apr 2022 to Nov 2022* **-** Micro Focus is one of the world's largest enterprise software providers delivering mission critical technology

- Spearheaded the integration of React into a mission critical legacy Java ecosystem creating a scalable i18n framework that enabled global localisation
- Provided mentorship and guidance to junior engineer on my team teaching how to do pull requests

WEB ENGINEER - VML Technology - *Nov 2019 to Mar 2022* **-** VML Technology pioneers new levels of video maximizing engagement through real time convergence of data

- Delivered core web platform features using React and Django, achieving 1:1 parity with mobile application for real time video engagement

TECHNICAL LEADERSHIP & COMMUNITY

- **Star Mentor**: Zero to Mastery is an online academy providing video courses on a wide range of programming topics with a Discord community of over 500,000 members. After 7 years of dedication towards learning and giving back to the community through mentorship of juniors I am now 1 of 20 people in the community that has been recognised and granted the title of Star Mentor

- **Tech Talks**: Passionate about education I have spoken at tech events in the data engineering community covering topics such as data manipulation with Apache Spark and how to create AI Skills

  - **AI Skills: What They Are and How to Create Your Own**: I presented a workshop teaching AI Skills. Showcasing what they are and how to create your own: https://github.com/markodowd/presentation-2026-08-29-ai-skills

- **Meetups:** Engaging and networking within the various tech communities is important in order to stay on top of current trends and as such I have attended ~30 different events within the Dublin area: **Meetup Profile**

- **Technical Blog:** I'm an avid book reader and have recently started doing a blog reviewing each book. The blog is available at **Personal Blog**

- **Open Source Contributions:** I believe in giving back to open source software when I identify an issue or some way in which I can contribute. Some examples of contributions:

  - **React Flow**: I noticed issues during development with the TypeScript typings in the library so provided a detailed explanation and discussion to fix: https://github.com/xyflow/web/issues/479

  - **TanStack Table:** When the Table project was early in development while reading the docs I identified issues and provided a detailed video which has since been completed: https://github.com/TanStack/table/issues/5193

CERTIFICATIONS

- AWS Certified Solutions Architect **- SAA-C03 -** *Amazon Web Services* **– Verify**
- AWS Certified AI Business Strategist **- AB1-C01 -** *Amazon Web Services* **– Verify**
- AWS Certified AI Practitioner **– AIF-C01 -** *Amazon Web Services* **– Verify**
- AWS Certified Cloud Practitioner **- CLF-C02 -** *Amazon Web Services* **– Verify**

EDUCATION

Queens University Belfast - BACHELOR OF SCIENCE (Bsc) MUSIC TECHNOLOGY